

Операционные системы

Поиск файлов. Перенаправление ввода-вывода. Просмотр запущенных процессов

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Цели и задачи работы

Процесс выполнения лабораторной работы

Выводы по проделанной работе

1. Цели и задачи работы

Ознакомление с файловой системой Linux, её структурой, именами и содержанием каталогов. Приобретение практических навыков по применению команд для работы с файлами и каталогами, по управлению процессами, по проверке использования диска и обслуживанию файловой системы.

- 1 Изучить перенаправление ввода-вывода
- 2 Изучить работу фильтров
- 3 Изучить команду поиска
- 4 Ознакомиться с управлением процессами
- 5 Ознакомиться с командами `df` `du`

2. Процесс выполнения лабораторной работы

```
miepihina@miepihina:~$ ls /etc/ > file.txt
miepihina@miepihina:~$ ls >> file.txt
miepihina@miepihina:~$ cat file.txt
abrt
adjtime
aliases
alsa
alternatives
anaconda
anacrontab
anthy-unicode.conf
asound.conf
at.deny
audit
authselect
avahi
bash_completion.d
bashrc
bindresvport.blacklist
binfmt.d
bluetooth
brlapi.key
brltty
brltty.conf
ceph
chromium
chrony.conf
```

Рисунок 1: Запись в файл

```
miepihina@miepihina:~$ grep .conf file.txt > conf.txt
miepihina@miepihina:~$ cat conf.txt
anthy-unicode.conf
asound.conf
brltty.conf
chrony.conf
cpupower-service.conf
dconf
dnsmasq.conf
dracut.conf
dracut.conf.d
fprindt.conf
fuse.conf
host.conf
idmapd.conf
kdump.conf
krb5.conf
krb5.conf.d
ld.so.conf
ld.so.conf.d
libaudit.conf
libuser.conf
locale.conf
logrotate.conf
makedumpfile.conf.sample
```

Рисунок 2: Поиск расширения .conf


```
/home/miepihina/work/study/2025-2026/Операционные системы/2026-1--study--os-intro/project-personal/stage05/report/_resources/csl
/home/miepihina/work/study/2025-2026/Операционные системы/2026-1--study--os-intro/project-personal/stage05/report/bib/cite.bib
/home/miepihina/work/study/2025-2026/Операционные системы/2026-1--study--os-intro/project-personal/stage06/report/_resources/csl
/home/miepihina/work/study/2025-2026/Операционные системы/2026-1--study--os-intro/project-personal/stage06/report/bib/cite.bib
/home/miepihina/work/study/2025-2026/Операционные системы/2026-1--study--os-intro/presentation/report/_resources/csl
/home/miepihina/work/study/2025-2026/Операционные системы/2026-1--study--os-intro/presentation/report/bib/cite.bib
/home/miepihina/git-extended/.git/hooks/commit-msg.sample
/home/miepihina/git-extended/.git/objects/7b/cc6afcd848e6c44670820ccebceb8266b85215
/home/miepihina/git-extended/.git/objects/c5
/home/miepihina/git-extended/.git/objects/ca
/home/miepihina/git-extended/.git/objects/ca/c45c97b0e6d4c05681e9ee45307c3e2611e901
/home/miepihina/git-extended/.git/objects/26/c03e290eb3ecab07642240e4153f3c57b0962c
/home/miepihina/git-extended/.git/objects/59/ca8fee3563a5661d44af3ded7acb92c521c4f4
/home/miepihina/git-extended/.git/objects/c0
/home/miepihina/git-extended/.git/config
/home/miepihina/conf.txt
miepihina@miepihina:~$
```

Рисунок 3: Поиск файлов

```
/etc/httpd
/etc/httpd/conf/httpd.conf
/etc/libibverbs.d/hfi1verbs.driver
/etc/libibverbs.d/hns.driver
find: '/etc/lvm/archive': Permission denied
/etc/logrotate.d/httpd
find: '/etc/lvm/backup': Permission denied
find: '/etc/lvm/cache': Permission denied
find: '/etc/lvm/devices': Permission denied
find: '/etc/nftables': Permission denied
find: '/etc/openvpn/client': Permission denied
/etc/nvme/hostnqn
/etc/nvme/hostid
find: '/etc/openvpn/server': Permission denied
find: '/etc/pki/rsyslog': Permission denied
find: '/etc/polkit-1/localauthority': Permission denied
find: '/etc/polkit-1/rules.d': Permission denied
find: '/etc/sos/cleaner': Permission denied
/etc/sane.d/dll.d/hpaio
/etc/sane.d/hp.conf
/etc/sane.d/hp3900.conf
/etc/sane.d/hp4200.conf
/etc/sane.d/hp5400.conf
/etc/sane.d/hpsj5s.conf
/etc/sane.d/hs2p.conf
:find: '/etc/ssh/sshd_config.d': Permission denied
find: '/etc/sssd': Permission denied
find: '/etc/sudoers.d': Permission denied
find: '/etc/userdb': Permission denied
```

Рисунок 4: Поиск файлов

```
miepihina@miepihina:~$  
miepihina@miepihina:~$ find /etc -name "h*" -print | less  
find: '/etc/audit': Permission denied  
find: '/etc/credstore.encrypted': Permission denied  
find: '/etc/credstore': Permission denied  
miepihina@miepihina:~$  
miepihina@miepihina:~$  
miepihina@miepihina:~$  
miepihina@miepihina:~$ find ~ -name "log*" > logfile &  
[1] 6418  
miepihina@miepihina:~$  
[1]+  Done                  find ~ -name "log*" > logfile  
miepihina@miepihina:~$ rm logfile  
miepihina@miepihina:~$
```

Рисунок 5: Фоновый запуск процесса

```
miepihina@miepihina:~$  
miepihina@miepihina:~$ find ~ -name "log*" > logfile &  
[1] 6418  
miepihina@miepihina:~$  
[1]+  Done                  find ~ -name "log*" > logfile  
miepihina@miepihina:~$ rm logfile  
miepihina@miepihina:~$  
miepihina@miepihina:~$  
miepihina@miepihina:~$ gedit &  
[1] 6456  
miepihina@miepihina:~$ ps | grep gedit  
    6456 pts/1    00:00:00 gedit  
miepihina@miepihina:~$ kill 6456  
[1]+  Terminated          gedit  
miepihina@miepihina:~$
```

Рисунок 6: Фоновый запуск и завершение процесса

```
DF(1)                                User Commands                                DF(1)

NAME
    df - report file system space usage

SYNOPSIS
    df [OPTION]... [FILE]...

DESCRIPTION
    This manual page documents the GNU version of df.  df displays the amount of space
    available on the file system containing each file name argument.  If no file name is
    given, the space available on all currently mounted file systems is shown.  Space is
    shown in 1K blocks by default, unless the environment variable POSIXLY_CORRECT is set,
    in which case 512-byte blocks are used.

    If an argument is the absolute file name of a device node containing a mounted file
    system, df shows the space available on that file system rather than on the file system
    containing the device node.  This version of df cannot show the space available on un-
    mounted file systems, because on most kinds of systems doing so requires non-portable
    intimate knowledge of file system structures.

OPTIONS
    Show information about the file system on which each FILE resides, or all file systems
    by default.

    Mandatory arguments to long options are mandatory for short options too.

    -a, --all
        include pseudo, duplicate, inaccessible file systems

    -B, --block-size=SIZE
        scale sizes by SIZE before printing them; e.g., '-BM' prints sizes in units of
        1,048,576 bytes; see SIZE format below

Manual page df(1) line 1 (press h for help or q to quit)
```

Рисунок 7: Справка по команде df

```
DU(1)                                User Commands                                DU(1)

NAME
    du - estimate file space usage

SYNOPSIS
    du [OPTION]... [FILE]...
    du [OPTION]... --files0-from=F

DESCRIPTION
    Summarize device usage of the set of FILES, recursively for directories.

    Mandatory arguments to long options are mandatory for short options too.

    -0, --null
        end each output line with NUL, not newline

    -a, --all
        write counts for all files, not just directories

    --apparent-size
        print apparent sizes rather than device usage; although the apparent size is
        usually smaller, it may be larger due to holes in ('sparse') files, internal
        fragmentation, indirect blocks, and the like

    -B, --block-size=SIZE
        scale sizes by SIZE before printing them; e.g., '-BM' prints sizes in units of
        1,048,576 bytes; see SIZE format below

    -b, --bytes
        equivalent to '--apparent-size --block-size=1'

    -c, --total
        Manual page du(1) line 1 (press h for help or q to quit)
```

Рисунок 8: Запуск команды df

```
miepihina@miepihina:~$ df
Filesystem      1K-blocks    Used Available Use% Mounted on
/dev/nvme0n1p3 155186176 91212516 63375148 60% /
devtmpfs         4011272      0    4011272  0% /dev
tmpfs            4035644      92   4035552  1% /dev/shm
tmpfs            1614260     1952   1612308  1% /run
tmpfs             1024        0      1024  0% /run/credentials/systemd-journald.service
tmpfs            4035648      60   4035588  1% /tmp
/dev/nvme0n1p3 155186176 91212516 63375148 60% /home
/dev/nvme0n1p2 1992552    518800   1352512 28% /boot
/dev/loop0       68480     68480      0 100% /var/lib/snapd/snap/core24/1349
/dev/loop2       49280     49280      0 100% /var/lib/snapd/snap/snapd/25935
/dev/loop1       115712    115712      0 100% /var/lib/snapd/snap/hugo/25676
tmpfs             1024        0      1024  0% /run/credentials/systemd-resolved.service
tmpfs            807128      168   806960  1% /run/user/1060
/dev/sr0         2677920    2677920      0 100% /run/media/miepihina/Fedora-WS-Live-43
miepihina@miepihina:~$
```

Рисунок 9: Справка по команде du

```
4      ./git-extended/.git/objects/6e
4      ./git-extended/.git/objects/c0
4      ./git-extended/.git/objects/27
4      ./git-extended/.git/objects/08
76     ./git-extended/.git/objects
8      ./git-extended/.git/logs/refs/heads
8      ./git-extended/.git/logs/refs/remotes/origin
8      ./git-extended/.git/logs/refs/remotes
16     ./git-extended/.git/logs/refs
20     ./git-extended/.git/logs
212    ./git-extended/.git
220    ./git-extended
0      ./monthly
0      ./reports/monthly/monthly
0      ./reports/monthly
0      ./reports
4      ./ski.plases/equipment
0      ./ski.plases/plans
4      ./ski.plases
0      ./australia
0      ./play/games/play
0      ./play/games
0      ./play
820720 .
miepihina@miepihina:~$
```

Рисунок 10: Запуск команды du


```
/home/miepihina/git-extended/.git/objects/dc
/home/miepihina/git-extended/.git/objects/59
/home/miepihina/git-extended/.git/objects/fc
/home/miepihina/git-extended/.git/objects/6e
/home/miepihina/git-extended/.git/objects/c0
/home/miepihina/git-extended/.git/objects/27
/home/miepihina/git-extended/.git/objects/08
/home/miepihina/git-extended/.git/logs
/home/miepihina/git-extended/.git/logs/refs
/home/miepihina/git-extended/.git/logs/refs/heads
/home/miepihina/git-extended/.git/logs/refs/remotes
/home/miepihina/git-extended/.git/logs/refs/remotes/origin
/home/miepihina/monthly
/home/miepihina/reports
/home/miepihina/reports/monthly
/home/miepihina/reports/monthly/monthly
/home/miepihina/ski.plases
/home/miepihina/ski.plases/equipment
/home/miepihina/ski.plases/plans
/home/miepihina/australia
/home/miepihina/play
/home/miepihina/play/games
/home/miepihina/play/games/play
miepihina@miepihina:~$
```

Рисунок 11: Поиск директорий

3. Выводы по проделанной работе

В данной работе мы ознакомились с инструментами поиска файлов и фильтрации текстовых данных. А также приобрели практические навыки по управлению процессами.